OSCILLATIONS

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. SIMPLE HARMONIC MOTION

Condition: $F = -kx$ or $a = -\omega^2 x$

$$x = A\sin(\omega t + \phi), \qquad \omega = 2\pi f = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$v = A\omega\cos(\omega t + \phi) = \omega\sqrt{A^2 - x^2}$$

$$a = -\omega^2 x, \quad v_{\max} = A\omega, \quad a_{\max} = A\omega^2$$

Phase relations: v leads x by $\frac{\pi}{2}$; a leads x by π

Graphs: v–x: ellipse; a–x: straight line, slope $-\omega^2$; v–a: ellipse

⋆ Common Pitfall

- At mean position: v max, $a = 0$; at extremes: $v = 0$, a max
- $x = A\sin\omega t + B\cos\omega t$ is SHM with amplitude $\sqrt{A^2 + B^2}$
- $x = A\sin^2\omega t$ is SHM with angular frequency 2ω, amplitude $A/2$ (about $A/2$)

2. TIME & PHASE SHORTCUTS

Motion (from mean or extreme)	Time
$0 \to A/2$	$T/12$
$0 \to A/\sqrt{2}$	$T/8$
$0 \to \sqrt{3}A/2$	$T/6$
$0 \to A$	$T/4$
$A \to A/2$	$T/6$
$A/2 \to -A/2$	$T/6$

Use the **phasor (reference circle)**: angle swept $= \omega t$

Average speed over full cycle $= \dfrac{4A}{T} = \dfrac{2}{\pi}v_{\max}$

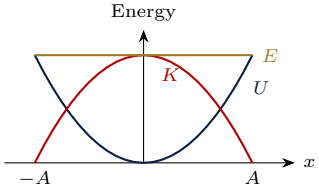
3. ENERGY IN SHM

$$K = \tfrac{1}{2}m\omega^2(A^2 - x^2), \qquad U = \tfrac{1}{2}m\omega^2 x^2$$

$$E = \tfrac{1}{2}m\omega^2 A^2 = \tfrac{1}{2}kA^2 = \text{constant}$$

$K = U$ at $x = \dfrac{A}{\sqrt{2}}$

Time-average: $\langle K\rangle = \langle U\rangle = \dfrac{E}{2}$; position-average: $\langle K\rangle = \dfrac{2E}{3}$, $\langle U\rangle = \dfrac{E}{3}$



⋆ Common Pitfall

- K and U oscillate with frequency $2f$ (period $T/2$)
- Total energy $\propto A^2\omega^2$

4. SPRING–MASS SYSTEMS

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Series: $\dfrac{1}{k} = \dfrac{1}{k_1} + \dfrac{1}{k_2}$ **Parallel:** $k = k_1 + k_2$

Cut spring: $kL = \text{const} \Rightarrow$ piece of length L/n has nk

Two masses on spring: $T = 2\pi\sqrt{\dfrac{\mu}{k}}$, $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$

Massive spring: $T = 2\pi\sqrt{\dfrac{m + m_s/3}{k}}$

Vertical spring: same T; equilibrium extension $x_0 = \dfrac{mg}{k} \Rightarrow T = 2\pi\sqrt{\dfrac{x_0}{g}}$

⋆ Common Pitfall

- T of spring–mass is independent of g (same on the moon, in a lift)
- Mass between two springs (both sides) → parallel combination

5. PENDULUMS

Simple pendulum: $T = 2\pi\sqrt{\dfrac{l}{g}}$ (small θ)

Effective g:

- Lift: $g_{\text{eff}} = g \pm a$ (up +, down −); free fall: $T \to \infty$
- Horizontal acceleration: $g_{\text{eff}} = \sqrt{g^2 + a^2}$
- Bob in liquid: $g_{\text{eff}} = g\left(1 - \dfrac{\rho_\ell}{\rho}\right)$
- Charged bob in field E: $\vec{g}_{\text{eff}} = \vec{g} + \dfrac{q\vec{E}}{m}$

Very long pendulum: $T = 2\pi\sqrt{\dfrac{1}{g\left(\frac{1}{l} + \frac{1}{R}\right)}}$, max ≈ 84.6 min

Seconds pendulum: $T = 2$ s, $l \approx 1$ m

Physical pendulum: $T = 2\pi\sqrt{\dfrac{I}{mgd}}$ (d = pivot to CM)

System	Period
Rod pivoted at end	$2\pi\sqrt{2L/3g}$
Ring on a nail	$2\pi\sqrt{2R/g}$
Disc pivoted at rim	$2\pi\sqrt{3R/2g}$
Torsion pendulum	$2\pi\sqrt{I/C}$

⋆ Common Pitfall

- T of simple pendulum is independent of mass and amplitude (small angles)
- Temperature rise: $\Delta T/T = \frac{1}{2}\alpha\Delta\theta$ (clock loses time)
- Use l to the **centre** of the bob

6. OTHER SHM SYSTEMS

System	Period
Liquid in U-tube (column L)	$2\pi\sqrt{L/2g}$
Floating cylinder (depth h)	$2\pi\sqrt{h/g}$
Tunnel through earth	$2\pi\sqrt{R/g} \approx 84.6$ min
Ball in bowl (radius R)	$2\pi\sqrt{R/g}$
Ball rolling in bowl	$2\pi\sqrt{7(R-r)/5g}$
Gas piston (isothermal)	$2\pi\sqrt{mV/PA^2}$

General method: displace by x, write restoring force $F = -k_{\text{eff}}x$, then $T = 2\pi\sqrt{m/k_{\text{eff}}}$
Or energy: $E = \frac{1}{2}m_{\text{eff}}\dot{x}^2 + \frac{1}{2}k_{\text{eff}}x^2$

⋆ Common Pitfall

- Tunnel along any chord through earth: same $T = 84.6$ min
- Adiabatic gas piston: replace P by γP

7. SUPERPOSITION OF SHMs

Same direction, same ω, phase difference ϕ:

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}, \quad \tan\delta = \frac{A_2\sin\phi}{A_1 + A_2\cos\phi}$$

Perpendicular SHMs (same ω): $\phi = 0, \pi \to$ line; $\phi = \pi/2 \to$ ellipse (circle if $A_1 = A_2$)

8. DAMPED & FORCED OSCILLATIONS

Damped: $F = -kx - bv$

$$x = A_0e^{-bt/2m}\cos(\omega' t + \phi), \qquad \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

Amplitude $A = A_0e^{-bt/2m}$, Energy $E = E_0e^{-bt/m}$

Forced: amplitude is maximum when driving $\omega_d \approx \omega_0$ (resonance); smaller damping $\Rightarrow$ sharper, taller peak

⋆ Common Pitfall

- Energy decays twice as fast as amplitude (time constant m/b vs $2m/b$)
- Damping lowers the frequency slightly: $\omega' < \omega_0$

9. JEE SHORTCUTS & MASTER TIPS

1. $v = \omega\sqrt{A^2 - x^2}$; $a = -\omega^2 x$
2. $K = U$ at $x = A/\sqrt{2}$; energy freq $= 2f$
3. $0 \to A/2$: $T/12$; $A \to A/2$: $T/6$
4. Series springs: $\frac{1}{k} = \sum\frac{1}{k_i}$; cut spring: $k \propto 1/L$
5. g_{eff} approach for every pendulum in a non-inertial frame
6. Physical pendulum: $T = 2\pi\sqrt{I/mgd}$

⋆ Master Confusion List

1. **Periodic $\neq$ SHM:** SHM needs $a \propto -x$
2. **Spring** T independent of g; pendulum T depends on g
3. **Series vs parallel springs:** opposite to resistors
4. **Time vs position average** of KE: $E/2$ vs $2E/3$
5. **Damped:** energy $\propto e^{-bt/m}$, amplitude $\propto e^{-bt/2m}$
6. $\sin^2\omega t$ motion has angular frequency 2ω

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